

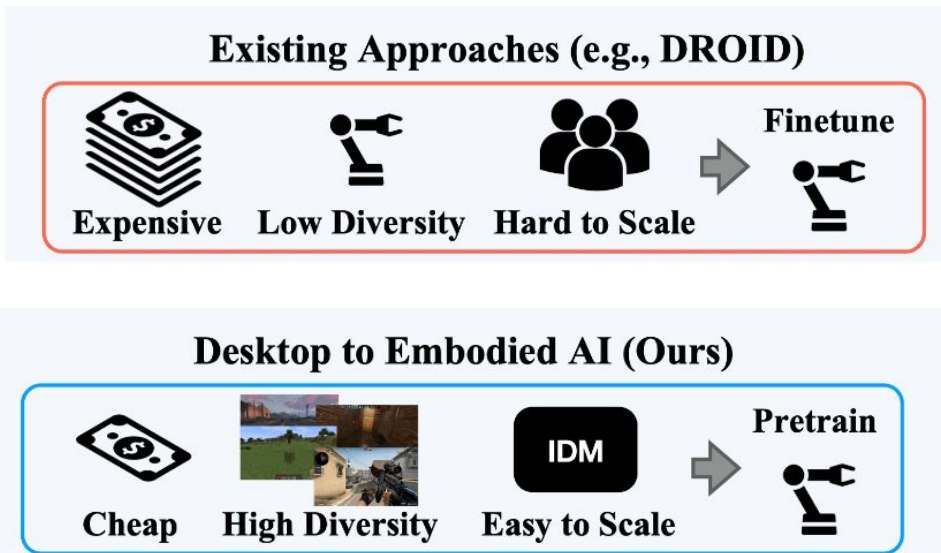
D2E: Scaling Vision-Action Pretraining on Desktop Data for Transfer to Embodied AI

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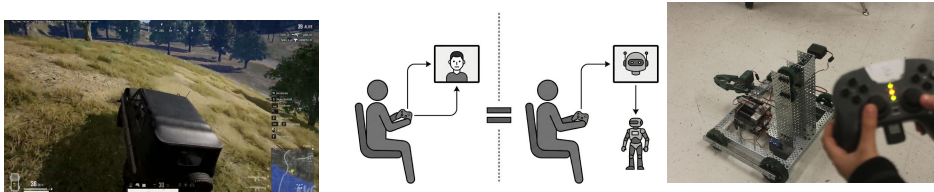
^{*} Indicates equal contribution
[†] Indicates co-corresponding

Why Desktop Data?

1. Data Scalability



2. Gaming Is Teleoperating the Protagonist



Why Transfer Works?

Action Modality Alignment

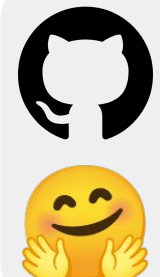
VAPT trains on vision-action trajectories
→ Learns how observations map to control

Goal-Directed Sequential Reasoning

Visual grounding + temporal planning
→ Same skills as robotic control

High Diversity

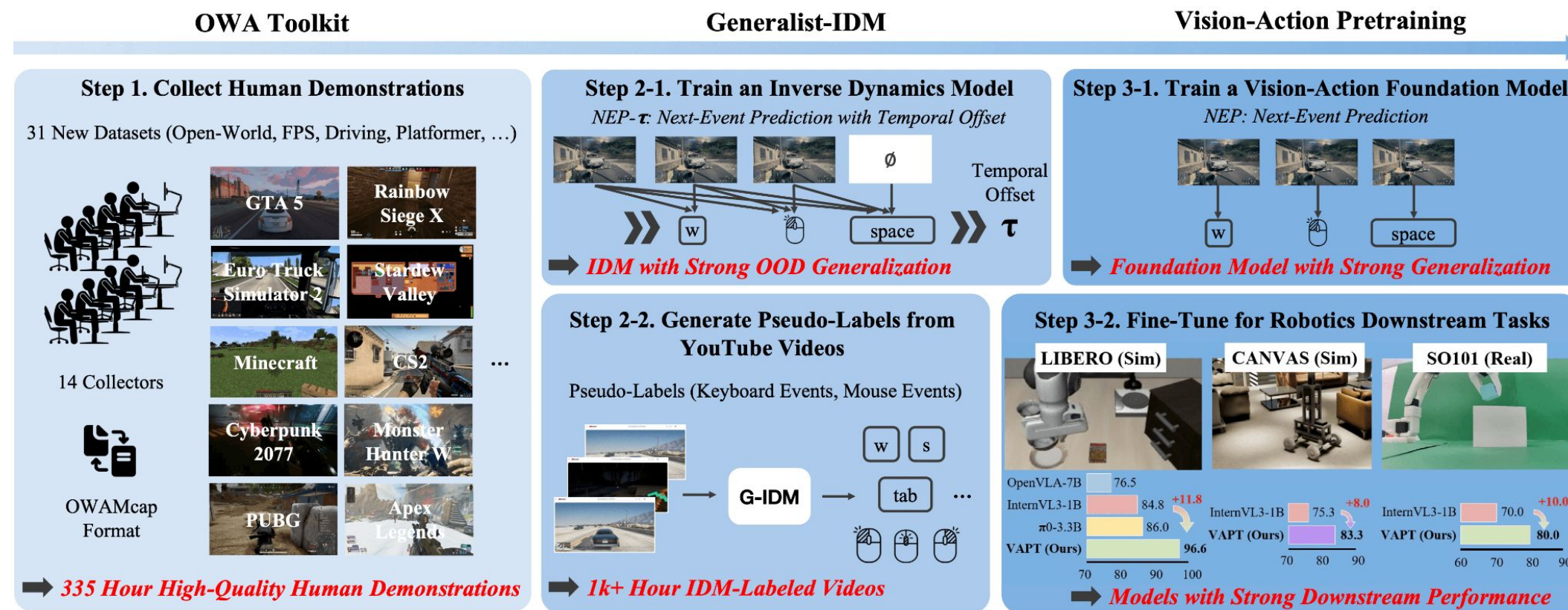
Games across 2D/3D, FPS/TPS/top-down
→ General-purpose control priors



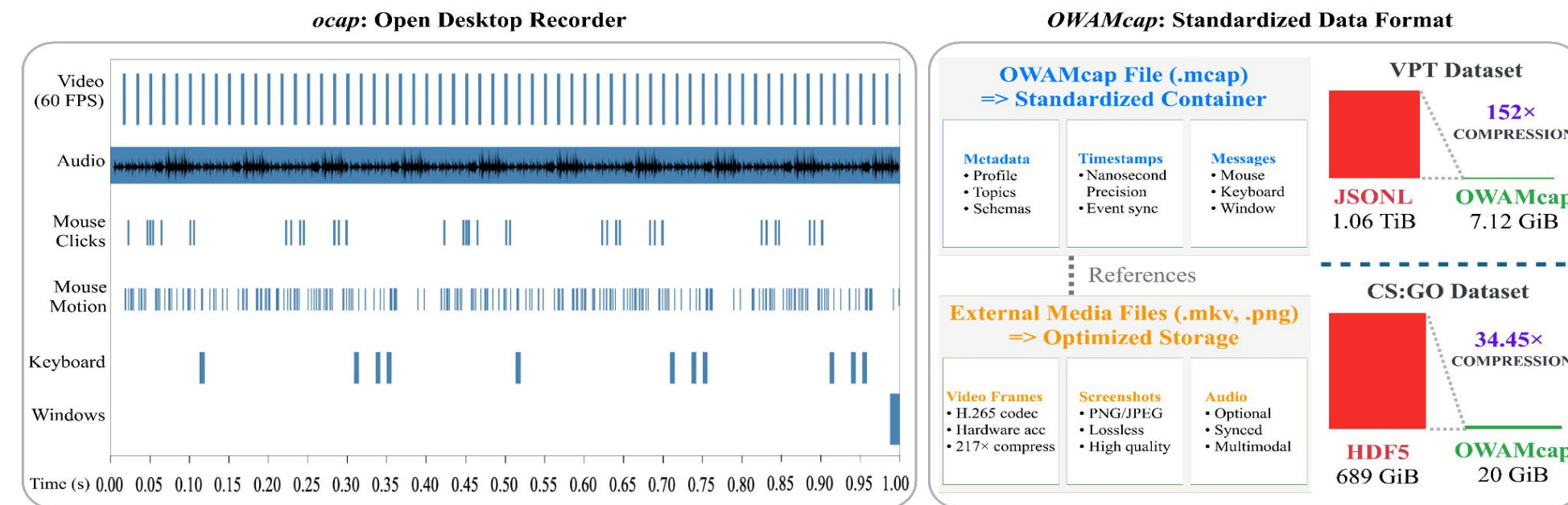
worv-ai.github.io/d2e/
Inference & evaluation code
Datasets + interactive viewer
Live demo on HF Spaces
Pretrained model weights

All resources (codes, models, datasets) are publicly available

Overview



OWA Toolkit



From screen to GPU: synchronized 60Hz capture, 152× compression, and 10× faster training

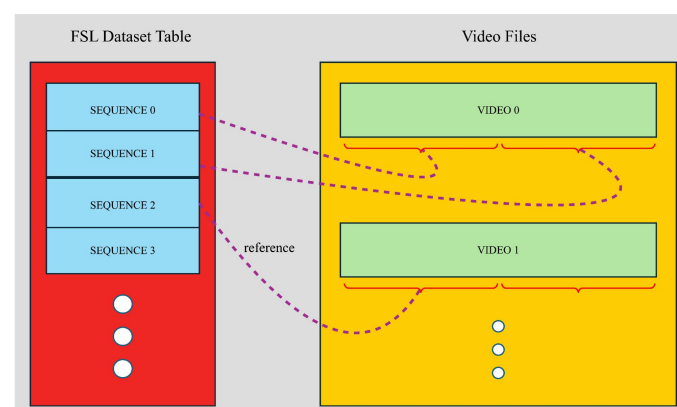
Data Pipeline

Configuration	Throughput (it/s)
Ours (1 worker)	4.77
Ours (4 workers)	4.73
Baseline (8 workers)	3.79
Baseline (12 workers)	4.42
Baseline (16 workers)	4.55

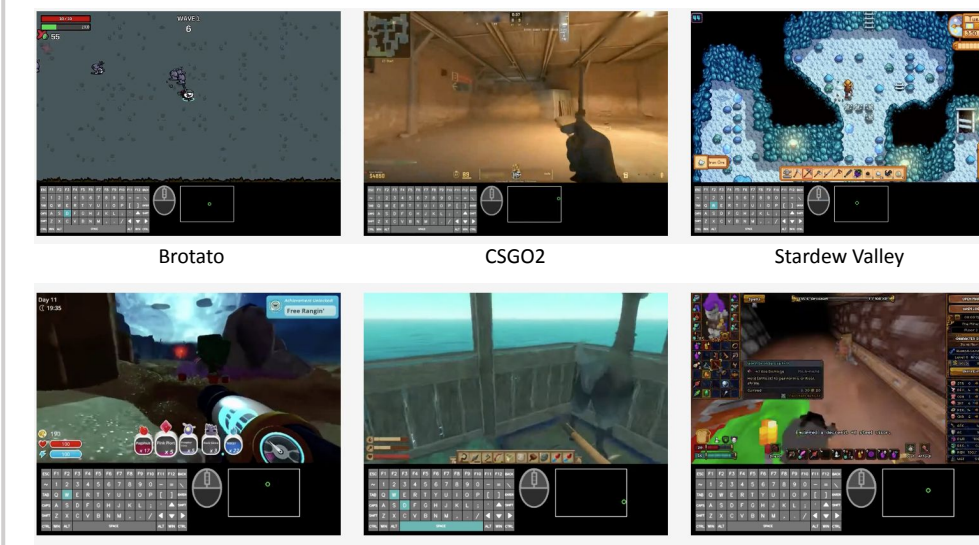
Training Throughput (InternVL3-1B)

Configuration	Throughput (img/s)	Avg. Read (KB/img)
Baseline	11.68	63.46
+ optimized x264	24.25	41.69
+ TorchCodec	79.73	770.39
+ Ours	119.16	18.73

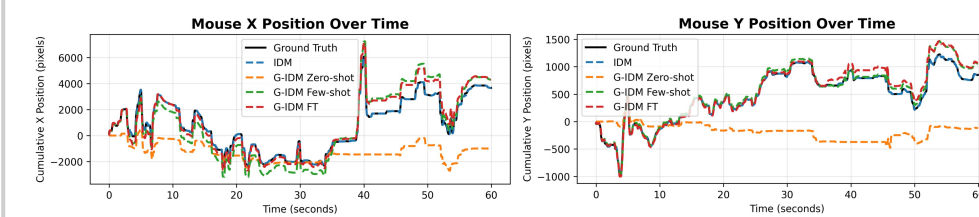
- Optimized x264 Parameters
- FSLDataset Construction
- Adaptive Batch Decoding



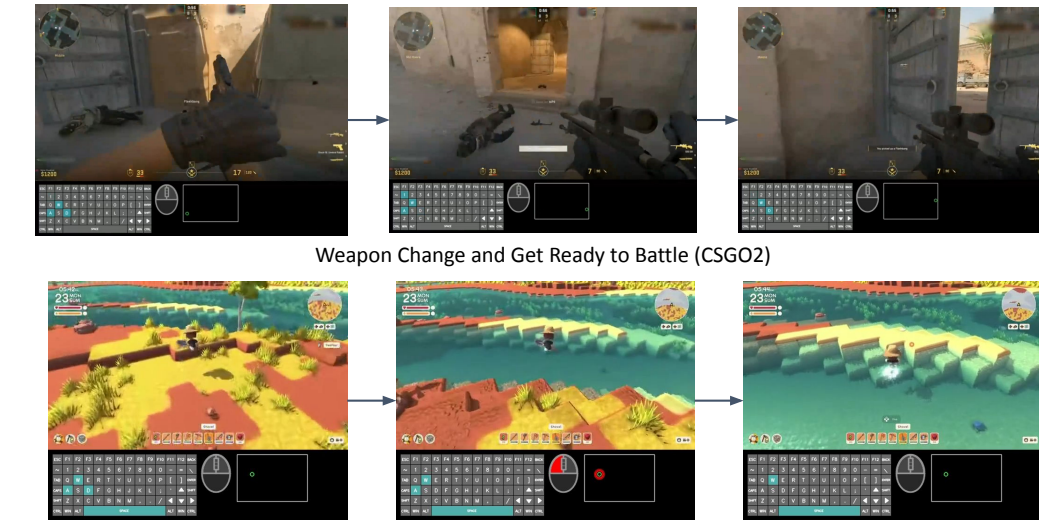
Generalist-IDM



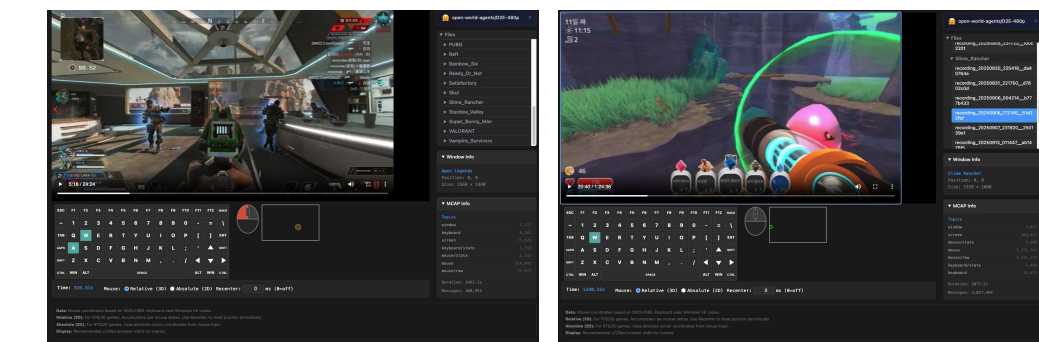
G-IDM Pseudo-Label Results on YouTube Game Videos



G-IDM Prediction Trajectory of Battlefield 6



Pseudo-Labeling Sequence Example from In-the-Wild YouTube



HF Dataset Visualizer (Publicly Available)

One model, all games:

G-IDM outperforms per-game specialists and pseudo-labels 1K+ hours of YouTube gameplay for ~\$800 (G-IDM training cost).

Evaluation Results of D2E

Method	Params	VLA Pt	Spatial	Object	Goal	10 (long)	Total
Octo (Octo Model Team et al., 2024)	93M	Yes	78.9	85.7	84.6	51.1	75.1
OpenVLA (Kim et al., 2024)	17B	Yes	84.7	88.4	79.2	53.7	76.5
DIT Policy (Dasari et al., 2025)	115M	No	84.2	96.3	85.4	63.8	82.4
70 Black et al. (2024)	3.3B	Yes	90.0	86.0	95.0	73.0	86.0
SmolVLA (Shukor et al., 2025)	2.25B	No	93.0	94.0	91.0	77.0	88.7
PI-KI (Driess et al., 2025)	300M	Yes	98.0	97.8	95.6	85.8	94.3
OpenVLA-OFT (Kim et al., 2025)	7B	Yes	97.6	98.4	97.9	94.5	97.1
Baseline (InternVL3-1B)	1B	No	94.4	97.0	93.6	54.2	84.8
+ VAPT w/o pseudo	1B	No	95.8	98.4	98.6	93.6	96.6
+ VAPT w/ pseudo	1B	No	89.6	98.2	93.8	87.2	92.2

LIBERO Manipulation Results

Method	Gallery	Office	Orchard	Street (road)	Street (side)	Total
	Mis. Prec.	Mis. Prec.	Mis. Prec.	Mis. Prec.	Mis. Prec.	
Baseline (InternVL3-1B)	53.3 100.0	100.0 100.0	53.3 40.0	94.4 100.0	40.0 73.3	75.3
+ VAPT w/o pseudo	33.3 93.3	93.3 100.0	53.3 53.3	88.9 91.7	53.3 93.3	75.3
+ VAPT w/ pseudo	53.3 93.3	100.0 100.0	86.7 60.0	88.9 100.0	73.3 80.0	83.3

CANVAS Navigation Results

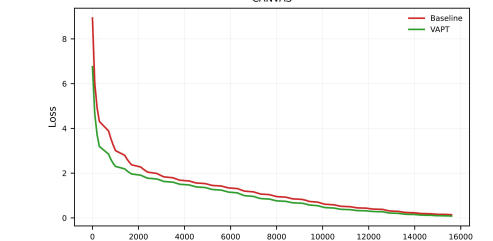
Method	Easy	Medium	Hard	Very Hard	Avg	Pick&Place
Baseline (InternVL3-1B)	55.4	14.5	1.7	8.0	19.9	70.0
+ VAPT w/o pseudo	53.6	18.2	8.3	20.0	25.0	80.0
+ VAPT w/ pseudo	52.1	16.4	6.7	24.0	24.8	80.0

Meta-World & SO101 Real-World Results

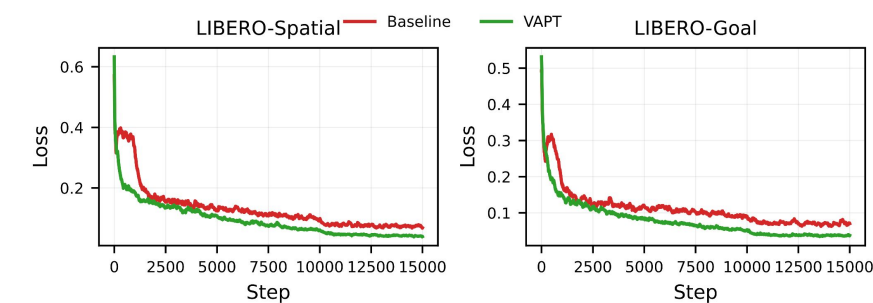
96.6% LIBERO, 83.3% CANVAS

1.3K hours of desktop gaming data pretrain a 1B model that matches 7× larger robotics models

Training Stability



Training Loss Curves for CANVAS (Navigation)



Training Loss Curves for LIBERO

More Stable Training

VAPT shows immediate convergence